

Homework 3- Solutions

Problem 1. Suppose $S = \{2, 3, 4, 5, 6, \dots\}$ and

$$P(k) = \frac{c}{3^k}, \quad k = 2, 3, 4, 5, 6, \dots$$

- (a) Find the value of c that makes this a valid probability distribution.
- (b) Find $P(\text{outcome is even})$.
- (c) Find $P(\text{outcome is less than or equal to } 6)$.

Show your work.

Solution.

- (a) To have a valid probability distribution we should have

$$1 = \sum_{k=2}^{\infty} P(k)$$

Hence,

$$\sum_{k=2}^{\infty} P(k) = \sum_{k=2}^{\infty} \frac{c}{3^k} = c \sum_{k=2}^{\infty} \frac{1}{3^k} = c \frac{\frac{1}{9}}{1 - \frac{1}{3}} = \frac{c}{6}$$

Hence,

$$\frac{c}{6} = 1$$

Hence, $c = 6$.

- (b) From Part (a), we know

$$P(k) = \frac{6}{3^k} = \frac{2}{3^{k-1}}, \quad k = 2, 3, 4, 5, 6, \dots$$

$$\begin{aligned}
P(\text{outcome is even}) &= P(2) + P(4) + P(6) + P(8) + \cdots = \\
&= \frac{2}{3} + \frac{2}{3^3} + \frac{2}{3^5} + \frac{2}{3^7} + \cdots \\
&= \frac{2}{3} \left(1 + \frac{1}{3^2} + \frac{1}{3^4} + \frac{1}{3^6} + \cdots \right) \\
&= \frac{2}{3} \left(1 + \frac{1}{9} + \frac{1}{81} + \frac{1}{729} + \cdots \right) \\
&= \frac{2}{3} \sum_{k=0}^{\infty} \left(\frac{1}{9} \right)^k = \frac{2}{3} \frac{1}{1 - \frac{1}{9}} = \frac{2}{3} \frac{9}{8} = \frac{3}{4}
\end{aligned}$$

(c)

$$\begin{aligned}
P(K \leq 6) &= P(2) + P(3) + P(4) + P(5) + P(6) \\
&= \frac{2}{3} + \frac{2}{3^2} + \frac{2}{3^3} + \frac{2}{3^4} + \frac{2}{3^5} \\
&= 2 \left(\frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \frac{1}{81} + \frac{1}{243} \right) \\
&= 2 \left(\frac{81 + 27 + 9 + 3 + 1}{243} \right) \\
&= 2 \frac{121}{243} = \frac{242}{243}
\end{aligned}$$

Problem 2. When an event A is independent from itself

- (a) Always
- (b) If and only if $P(A) = 1$
- (c) If and only if $P(A) = 0$
- (d) If and only if $P(A)$ is either 0 or 1.

Show your work. If A is independent from itself then,

$$P(A \cap A) = P(A).P(A)$$

But $A \cap A = A$, hence,

$$P(A) = P(A).P(A)$$

Hence,

$$P(A).(P(A) - 1) = 0$$

Hence $P(A) = 0$ or $P(A) = 1$. Hence, $A = \phi$ or $A = S$. Hence, (d) is the only correct option.

Problem 3. Let event M = taking a math class.

Let event S = taking a science class.

Then, M and S = taking a math class and a science class.

Suppose $P(M) = 0.4$, $P(S) = 0.6$, and $P(M \text{ AND } S) = 0.3$.

Are M and S independent?

☐ No

☐ Yes

Show your work.

Solution.

$$P(M) \cdot P(S) = 0.4 \cdot 0.6 = 0.24 \neq P(M \cap S)$$

Since $P(M) \cdot P(S) \neq P(M \cap S)$, M and S are not independent.

Problem 4. A player throws darts at a target. On each trial, independently of the other trials, he hits the bull's-eye with probability .06. How many times should he throw so that his probability of hitting the bull's-eye at least once is 0.5?

Solution.

S = Success = Hitting the bull's-eye

F = Failure = Not hitting the bull's-eye

$P(S_i) = 0.06$ and $P(F_i) = 1 - 0.06 = 0.94$.

n = total number of trials

$$P(\text{at least one success in } n \text{ trials}) = 1 - P(\text{no success in } n \text{ trials})$$

Since events are independent,

$$P(\text{at least one success in } n \text{ trials}) = 1 - P(F_1 \cap F_2 \cap F_3 \cap \cdots \cap F_n)$$

Where F_i = failure in the i th trial. $P(F_i) = 0.94$ for every i .

$$P(\text{at least one success in } n \text{ trials}) = 1 - P(F_1) \cdot P(F_2) \cdot P(F_3) \cdots P(F_n) = 1 - (0.94)^n$$

We should find n such that

$$P(\text{at least one success in } n \text{ trials}) = 1 - (0.94)^n = 0.5$$

Hence,

$$(0.94)^n = 0.5$$

Hence,

$$n = \frac{\ln 0.5}{\ln 0.94}$$

(This line is for your knowledge and it has no rule in your grade, $n = 11.202$. Approximately n can be estimated as 11.)

Problem 5. Two fair six-sided dice are tossed independently. Let X = the maximum of the two tosses.

- What is the pmf of X ?
- Determine the cdf of X and graph it.

Solution.

(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

X = maximum of the two dice.

X can take values of 1, 2, 3, 4, 5, 6.

The tabular representation of pmf X will be

x	1	2	3	4	5	6
$p_X(x)$	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{5}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	$\frac{11}{36}$

$$F_X(x) = \begin{cases} 0 & x < 1 \\ \frac{1}{36} & 1 \leq x < 2 \\ \frac{4}{36} & 2 \leq x < 3 \\ \frac{9}{36} & 3 \leq x < 4 \\ \frac{16}{36} & 4 \leq x < 5 \\ \frac{25}{36} & 5 \leq x < 6 \\ 1 & x \geq 6 \end{cases}$$

Problem 6. Assume that CDF of a discrete random variable X is given as follows.

$$F(x) = \begin{cases} 0 & x < 1 \\ .30 & 1 \leq x < 3 \\ .40 & 3 \leq x < 4 \\ .45 & 4 \leq x < 6 \\ .60 & 6 \leq x < 12 \\ 1 & 12 \leq x \end{cases}$$

- a. What is the pmf of X?
- b. Using just the cdf, compute $P(3 \leq X \leq 6)$ and $P(4 \leq X)$.

Solution. CDF $F_X(x)$ is a step function, hence X is a discrete random variable. X can take values of 1, 3, 4, 6, 12.

The tabular representation of pmf of X will be

x	1	3	4	6	12
$p_X(x)$	0.3	0.4-0.3	0.45-0.4	0.6-0.45	1-0.6

Hence,

x	1	3	4	6	12
$p_X(x)$	0.3	0.1	0.05	0.15	0.4

$$\begin{aligned} \text{(b) } P(3 \leq X \leq 6) &= F(6) - F(3^-) = F(6) - \lim_{x \rightarrow 3^-} F(x) = 0.6 - 0.3 = 0.3 \\ P(X \geq 4) &= 1 - P(X < 4) = 1 - F(4^-) = 1 - \lim_{x \rightarrow 4^-} F(x) = 1 - 0.4 = 0.6 \end{aligned}$$